BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021 • Paper Code: PR IX (01) 21

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** Biology Paper–IX (01) (21)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. The study of the use of organisms for the benefits and welfare of humanity is called.....

(A) Biotechnology
(B) Genetics
(C) Taxonomy
(D) Physiology

Q2. The first scientific classification was proposed by a Swiss scientist named:

(A) Gregor John Mendel
(B) Carolus Linnaes
(C) Charles Darwin
(D) Ernst Haeckel

Q3. The diffusion of solvent through a semi permeable membrane is called:

(A) Osmosis
(B) Facilitated diffusion
(C) Exocytosis
(D) Endocytosis

Q4. .....tissue is responsible for the transport of water and dissolved substances.

(A) Phloem
(B) Xylem
(C) Collenchyma
(D) Sclerenchyma

Q5. The final event in interphase occurs with the replication of:

(A) DNA
(B) Chromosomes
(C) Centrioles
(D) Genes

Q6. Crossing over occurs during:

(A) Prophase – I
(B) Ana Phase – I
(C) Meta Phase–II
(D) Telophase – II

Q7. The breakdown of larger molecules into smaller ones is known as:

(A) Anabolism
(B) Catabolism
(C) Metabolism
(D) Canabolism

Q8. .....is the study of transformation of energy in organisms.

(A) Photosynthesis
(B) Respiration
(C) Bioenergetics
(D) Transpiration

Q9. Glycolysis takes place in:

(A) Nucleus
(B) Mitochondria
(C) Golgi bodies
(D) Cytoplasm

Q10. One gram of carbohydrate (glucose) when fully oxidized produces:

(A) 2 k.cal
(B) 3 k.cal
(C) 4 k.cal
(D) 9 k.cal

Q11. Air in motion is called:

(A) Atmosphere
(B) Wind
(C) Humidity
(D) Moisture

Q12. Red blood cells are called:

(A) Erythrocytes
(B) Leucocytes
(C) Thrombocytes
(D) Phagocytes

Section-B

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 32

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

QII. Answer any EIGHT parts. Each part carries FOUR marks.

1. What are Bioelements? Give the percentage composition (by mass) of bioelements in organisms.

2. What are AFA king observations? Give AFA king hypothesis about his observations.

3. What is endoplasmic reticulum? Differentiate between smooth and rough endoplasmic reticulum.

4. Explain Active and Passive transport with examples.

5. What is cytokinesis? How this process occur in plant cell?
6. What is metabolism? Differentiate between Anabolism and Catabolism.
7. Briefly describe light reaction as a step of mechanism of photosynthesis.
8. What is aerobic respiration? Name its 3 main steps.
9. What is status of virus in classification?
10. What is transpiration? Name the factors affecting the rate of transpiration.
11. Write short note on Erythrocytes.

Section-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 21

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII.

(a) [3] Briefly describe observations about the causes of malaria.

(b) [4] What are muscular tissues? Briefly describe their types.

QIV.

(a) [3] Briefly describe limitations of two kingdom system of classification.

(b) [4] Write short note on Krebs Cycle.

QV.

(a) [3] Briefly describe lock and key model of enzyme action.

(b) [4] Briefly describe the significance of meiosis.

QVI.

(a) [3] Write short note on fats.

(b) [4] Briefly describe and draw the internal structure of human heart.